1 I

1.1 (1)

The susceptibility per spin is defined as

$$\chi = \frac{1}{N}\frac{\partial \langle M_x\rangle_h}{\partial h}\bigg|_{h=0}, \qquad \langle M_x\rangle_h = \frac{\mathrm{Tr}(e^{-\beta H}M_x)}{\mathrm{Tr}(e^{-\beta H})}.$$

Since $H = -hM_x + H_0$, we have

$$\frac{\partial e^{-\beta H}}{\partial h} = \beta M_x e^{-\beta H}.$$

Differentiating the thermal average using the quotient rule gives

$$\begin{aligned}\frac{\partial \langle M_x\rangle_h}{\partial h} &= \frac{\beta\,\mathrm{Tr}(e^{-\beta H}M_x^2)}{\mathrm{Tr}(e^{-\beta H})} - \frac{\beta\,\mathrm{Tr}(e^{-\beta H}M_x)^2}{[\mathrm{Tr}(e^{-\beta H})]^2} \\ &= \beta\big(\langle M_x^2\rangle_h - \langle M_x\rangle_h^2\big).\end{aligned}$$

Therefore,

$$\boxed{\chi = \frac{\beta}{N}\left(\langle M_x^2\rangle_0 - \langle M_x\rangle_0^2\right)}.$$

1.2 (2)

From $M_x = \sum_{i=1}^N S_i^x$,

$$\langle M_x^2\rangle_0 = \left\langle \sum_{i=1}^N S_i^x \sum_{j=1}^N S_j^x \right\rangle_0 = \sum_{i,j=1}^N \langle S_i^x S_j^x\rangle_0,$$

$$\langle M_x\rangle_0^2 = \sum_{i,j=1}^N \langle S_i^x\rangle_0 \langle S_j^x\rangle_0.$$

Thus, using $C_{i,j} = \langle S_i^x S_j^x\rangle_0 - \langle S_i^x\rangle_0\langle S_j^x\rangle_0$,

$$\boxed{\langle M_x^2\rangle_0 - \langle M_x\rangle_0^2 = \sum_{i,j=1}^N C_{i,j}}.$$

2 II

2.1 (3)

When $H_0 = 0$ and $h = 0$, we have $H = 0$. The spin angles are independent and uniformly distributed over $[-\pi, \pi)$. For $i \neq j$,

$$\langle S_i^x S_j^x\rangle_0 = \langle \cos\theta_i\rangle_0 \langle \cos\theta_j\rangle_0 = 0,$$

since

$$\langle S_i^x\rangle_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos\theta_i \, d\theta_i = 0.$$

For $i = j$,

$$\langle (S_i^x)^2\rangle_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos^2\theta_i \, d\theta_i = \frac{1}{2}.$$

Consequently,

$$\boxed{C_{i,j} = \frac{1}{2}\delta_{ij}}.$$

2.2 (4)

Substituting the result of (3) into (1)–(2),

$$\chi = \frac{\beta}{N}\sum_{i,j=1}^{N} C_{i,j} = \frac{\beta}{N}\cdot\frac{N}{2}.$$

Hence,

$$\boxed{\chi = \frac{\beta}{2} = \frac{1}{2k_BT} \propto \frac{1}{T}},$$

which is Curie's law.

3 III

3.1 (5)

For the one-dimensional chain with S_0 fixed,

$$H_0 = -J\sum_{i=1}^{N} \boldsymbol{S}_{i-1}\cdot\boldsymbol{S}_i = -J\sum_{i=1}^{N}\cos\phi_i.$$

The partition function factorizes:

$$\begin{aligned} Z = \mathrm{Tr}(e^{-\beta H_0}) &= \frac{1}{(2\pi)^N}\int_{-\pi}^{\pi} d\phi_1 \cdots \int_{-\pi}^{\pi} d\phi_N \exp\left(\beta J\sum_{i=1}^{N}\cos\phi_i\right) \\ &= \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\beta J\cos\phi}\, d\phi\right]^N = \boxed{\left[\frac{I(\beta J)}{2\pi}\right]^N}, \end{aligned}$$

where $I(x) = \int_{-\pi}^{\pi} e^{x\cos\phi}\, d\phi$, as defined in the problem. (This is the function denoted by $Z(x)$ in the scratch work.)

The Helmholtz free energy and internal energy are

$$\boxed{F = -Nk_BT\ln\left[\frac{I(\beta J)}{2\pi}\right]},$$

$$\begin{aligned} U = -\frac{\partial \ln Z}{\partial\beta} &= -N\frac{\partial}{\partial\beta}\ln I(\beta J) \\ &= \boxed{-NJ\frac{I'(\beta J)}{I(\beta J)}}. \end{aligned}$$

3.2 (6)

For high temperatures, $x = \beta J \ll 1$. Expanding the exponential,

$$\begin{aligned} I(x) &= \int_{-\pi}^{\pi} e^{x\cos\phi}\, d\phi \\ &\simeq \int_{-\pi}^{\pi} \left(1 + x\cos\phi + \frac{x^2}{2}\cos^2\phi\right) d\phi \\ &= 2\pi + \frac{\pi x^2}{2} = 2\pi\left(1 + \frac{x^2}{4}\right). \end{aligned}$$

Therefore, $I'(x) \simeq \pi x$, and

$$\begin{aligned} U &= -NJ\frac{I'(\beta J)}{I(\beta J)} \simeq -NJ\frac{\pi\beta J}{2\pi} = -\frac{NJ^2}{2k_BT}, \\ C &= \frac{\partial U}{\partial T} \simeq \frac{NJ^2}{2k_BT^2}. \end{aligned}$$

Thus,

$$\boxed{U \simeq -\frac{NJ^2}{2k_BT}, \qquad C \simeq \frac{NJ^2}{2k_BT^2}}.$$

3.3 (7)

For low temperatures, $x = \beta J \gg 1$. The main contribution to $I(x)$ comes from $\phi \approx 0$, where

$$\cos\phi \simeq 1 - \frac{\phi^2}{2}.$$

Using the Gaussian integral,

$$I(x) \simeq e^x \int_{-\infty}^{+\infty} e^{-x\phi^2/2}\, d\phi = e^x\sqrt{\frac{2\pi}{x}}.$$

Hence,

$$\ln I(x) \simeq x + \frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln x, \qquad \frac{I'(x)}{I(x)} \simeq 1 - \frac{1}{2x}.$$

The internal energy becomes

$$\begin{aligned} U &= -NJ\frac{I'(\beta J)}{I(\beta J)} \simeq -NJ\left(1 - \frac{1}{2\beta J}\right) \\ &= -NJ + \frac{N}{2}k_BT. \end{aligned}$$

Therefore,

$$\boxed{U \simeq -NJ + \frac{N}{2}k_BT, \qquad C \simeq \frac{N}{2}k_B}.$$

4 IV

4.1 (8)

When S_0 is not fixed, rotational symmetry gives $\langle \boldsymbol{S}_i\rangle_0 = 0$ and

$$C_{l,0} = \frac{1}{2}\langle \boldsymbol{S}_l \cdot \boldsymbol{S}_0\rangle_0 = \frac{1}{2}\langle\cos(\theta_l - \theta_0)\rangle_0.$$

Since $\theta_{l+1} - \theta_0 = (\theta_l - \theta_0) + \phi_{l+1}$,

$$\begin{aligned} C_{l+1,0} &= \frac{1}{2}\left\langle \cos\big[(\theta_l - \theta_0) + \phi_{l+1}\big]\right\rangle_0 \\ &= \frac{1}{2}\big\langle \cos(\theta_l - \theta_0)\cos\phi_{l+1}\big\rangle_0 \\ &\quad - \frac{1}{2}\big\langle \sin(\theta_l - \theta_0)\sin\phi_{l+1}\big\rangle_0. \end{aligned}$$

The relative angle ϕ_{l+1} is independent of the preceding ones, and $\langle \sin\phi_{l+1}\rangle_0 = 0$ by symmetry. Thus,

$$C_{l+1,0} = \langle \cos\phi_{l+1}\rangle_0 C_{l,0}.$$

Moreover,

$$\langle \cos\phi\rangle_0 = \frac{\int_{-\pi}^{\pi} \cos\phi\, e^{\beta J\cos\phi}\, d\phi}{\int_{-\pi}^{\pi} e^{\beta J\cos\phi}\, d\phi} = \frac{I'(\beta J)}{I(\beta J)}.$$

Therefore,

$$\boxed{C_{l+1,0} = \frac{I'(\beta J)}{I(\beta J)} C_{l,0}}.$$

Since $C_{0,0} = \frac{1}{2}$, iteration gives

$$C_{l,0} = \frac{1}{2}\left[\frac{I'(\beta J)}{I(\beta J)}\right]^l.$$

Comparing this with $C_{l,0} = \frac{1}{2}e^{-l/\xi}$, we obtain

$$\boxed{\xi = \left[\ln\frac{I(\beta J)}{I'(\beta J)}\right]^{-1}}.$$

Here the distance l is measured in lattice spacings.